State Machine Tea Brewer with Blynk

Assignment 8 – State Machine Tea Brewer with Blynk

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Goals

- Create and navigate state machines
- Connect speaker and play tones
- Build Blynk app to control interface

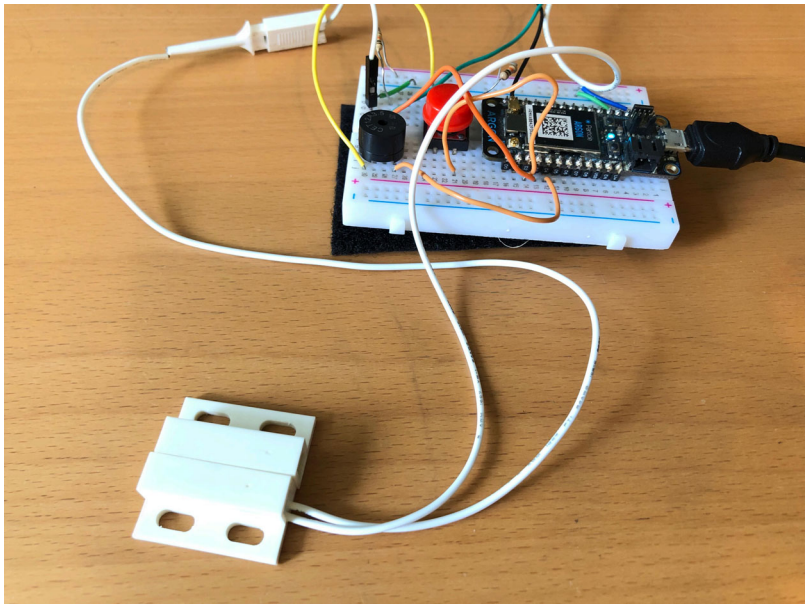
Overview

This assignment is to create a device that simulates brewing tea using state machines and a Blynk interface. The state machine machine logic will control how the device moves through the tea brewing states, and feedback and control will be available through the Blynk app. When brewing is finished, a song will optionally play on the Photon 2 through a speaker.

Here is a video overview: https://youtu.be/lqQhFIVWDR0 (https://youtu.be/lqQhFIVWDR0)

Note: Although an older microcontroller is used in the video, the assignment is still the same for the Photon 2.

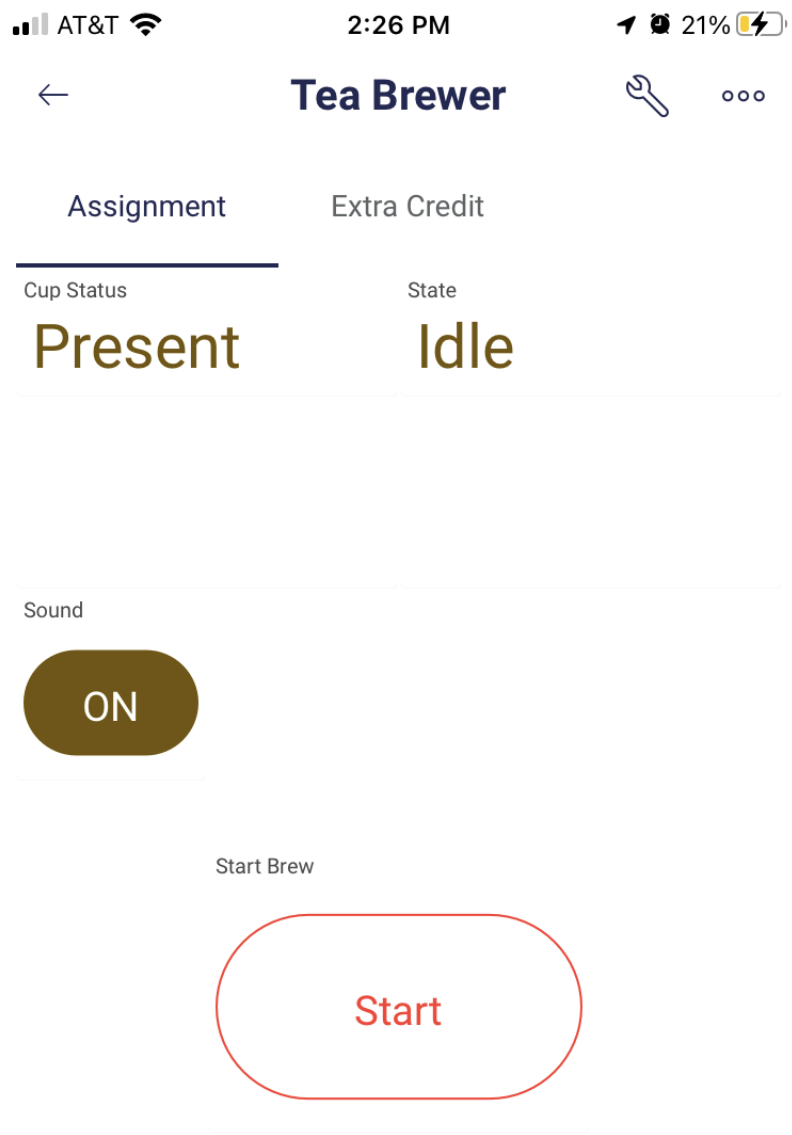
Device



This diagram uses an older microcontroller and may not represent the exact wiring you use in your assignment.

Note: The red button in the photo was used for testing only. You don't need to include it or use it in your assignment.

Blynk App


AT&T
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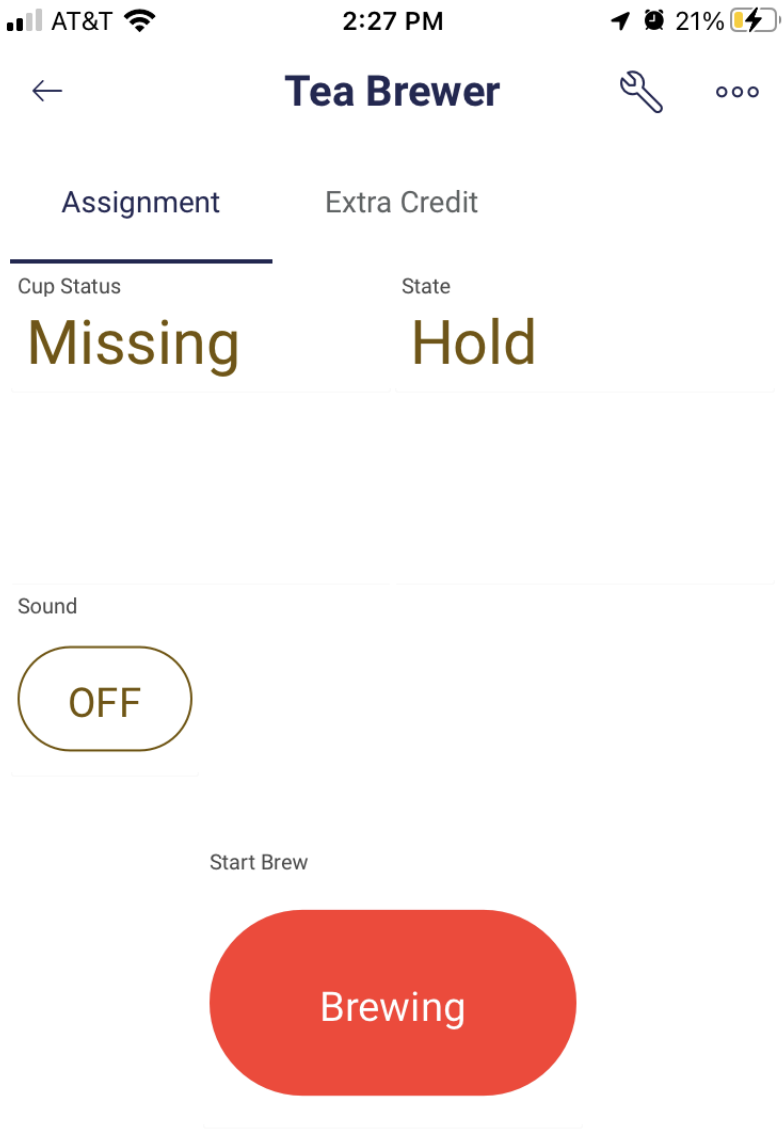
Tea Brewer
Assignment
Extra Credit
Cup Status
Present
State
Idle
Sound
ON
Start Brew
Start



Components

- Photon 2
- Breadboard
- 1 x buzzer
- 1 x magnetic switch

State Machine Logic

There are 5 states in the brewing process. Note that we will use **seconds** in places of **minutes**

1. **Idle**: nothing is happening until *Start button* on Blynk app is pressed to start brewing
2. **Heat**: water is heating up (takes 6 seconds)
3. **Steep**: tea is submerged in water (takes 3 seconds)
4. **Hold**: intermediate stage between **Steep** and **Pour**. Allows device to make sure magnetic switch is closed. If switch is open, device stays in **Hold**
5. **Pour**: liquid pours into cup (takes 4 seconds)

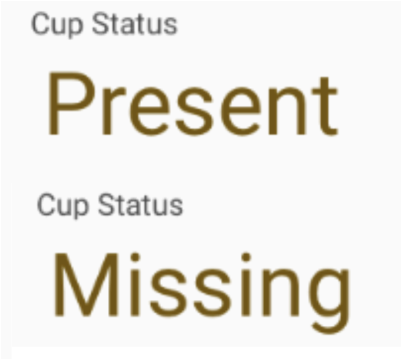
State Transition Table

State	Duration	Input	Next State
Idle	indefinite	Brew_Active == true	Heat
Heat	6 sec	Brew_Active == false	Idle
Heat	6 sec	(Brew_Active == true) and (Heat_Time >= Goal_Heat_Time)	Steep
Steep	3 sec	Brew_Active == false	Idle
Steep	3 sec	(Brew_Active == true) and (Brew_Time >= Goal_Brew_Time)	Hold
Hold	indefinite	Brew_Active == false	Idle
Hold	indefinite	((Brew_Active == true) and (Cup_Present == false)	Hold
Hold	indefinite	(Brew_Active == true) and (Cup_Present == true)	Pour
Pour	4 sec	Brew_Active == false	Idle
Pour	4 sec	(Brew_Active == true) and (Pour_Time >= Goal_Pour_Time)	Idle

Notes

- `Brew_Active` is based on the Blynk *Start button*
- `Cup_Present` is based on the magnetic switch

App Functionality

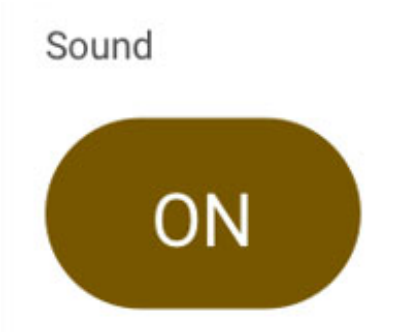


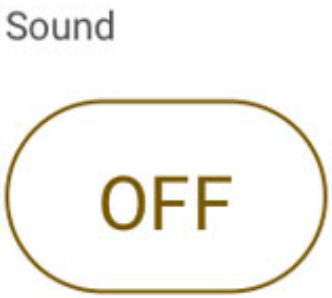
Cup Status

- Displays if the magnetic switch status
    - Switch closed means cup is present and brewing can continue
    - Switch open means cup is missing and pouring should not proceed

State status

- Displays name of current brew state





Sound start / stop button

- Pressing on will enable a song to play at the end of the **Pour** state
- Pressing off will disable song
- This should be configured as a **switch** in the Blynk app

Brewing start / stop button

- Pressing start will start the brewing process. When brewing process is complete, button will switch to inactive
- Pressing button during brew process will automatically stop brewing
- This should be configured as a **switch** in the Blynk app

Blynk Datastreams

Name	Pin	Type	Values	Data FROM	Data TO
`String Cup Status`	`V2`	String	`Present` , `Missing`	Photon 2	app
`Button Start Brew`	`V3`	integer	`0` , `1`	app	Photon 2
`String State`	`V4`	String	`Idle` , `Heat` , `Steep` , `Hold` , `Pour`	Photon 2	app
`Button Sound`	`V5`	integer	`0` , `1`	app	Photon 2

Requirements

- Create a Fritzing breadboard prototype of your design. No sample is provided so it is up to you to layout the components
- Once you are are finished, build the device
- It is recommended to code the device in stages, and test at each stage

Stage 1: Create State Machine States and Transition Logic

- Create an `enum` to represent the 5 states
- Create `updateNextState()` function with conditional logic to transition to different states. Transitions will be based on a combination of
  - current brew state (**Idle**, **Heat**, **Steep**, **Hold**, **Pour**)
  - time in current brew state
  - whether or not the brew process is active or not
  - magnetic switch state
- Within `updateNextState()` set the appropriate duration for the next state every time a transition is about to happen

Testing Stage 1

- Using Serial print statements, test that device progresses through all states with no timers. Don't worry about Blynk interface yet–manually set brewing process to be active
- Then test if device can progress through states with correct timing
- Finally test magnetic switch to make sure that device stays in **Hold** stage if cup is missing

Stage 2: Connect Blynk, Set Up Datastreams, and Design Blynk App

- Install the `blynk` library in your code
- Create a Blynk project and set up the required datastreams based on the table above
- Connect your code to the Blynk template
- Using the guide above, create the appropriate app elements to allow for user interaction
- Connect each screen element to the appropriate datastream

Testing Stage 2

- Flash your project, and open the Blynk app. Make sure that the blank template opens

Stage 3: Enable communication between Blynk app and Photon 2

- There are two app elements that send information from the app to the Photon 2
  - `Button Start Brew`
  - `Button Sound`
- Write the code that receives the infomation from the app and updates your code variables as necessary

Testing stage 3

- Use Serial print statements to make sure that values are being updated properly
- Make sure that your device stays idle until start button is pressed. Once it is pressed, device should move through each state. If button is pressed while brewing is happening, brewing should stop and device should return to **Idle**

Stage 4: Enable communication between Photon 2 and app

- There are two app elements that receive information from the app
  - `String State`
  - `String Cup Status`
- Write the code that the sends the appropriate infomation to the app
- Remember that all updates FROM Photon 2 TO app should take place in the `loop()` function inside a `millis()` block

Testing Stage 4

- Make sure that state label changes as device moves through states
- Make sure that cup status updates as magnetic switch is opened and closed

Stage 5: Play Song

- Create a melody that will play when the **Pour** state finishes. Melody should at least 16 notes. You can use the list of musical notes and frequencies (or do something different).
- Recommendation: Store melody notes in a array such as `int melodyNotes[16]` and note durations in array such as `int melodyNotesDuration[16]`. Note that `16` is just an example number of notes. You have have more notes in your song

Testing Stage 5

- Make sure that when **Pour** state finishes, your melody plays
- Make sure that Blynk app button enables / disables a variable that allows song to play or not

**Naming convention **

(replace # with the current assignment number)

- **Project Name**: `tac348_a#_lastname_firstname`
- **Zip File** (project folder): `tac348_a#_lastname_firstname.zip`

Video Submission Checklist

To receive credit, your submitted video must clearly demonstrate the following:

- [ ] Pressing *Start button* on Blynk app (and it functions properly)
- [ ] States transitioning
- [ ] Pressing *Sound* on / off button* on Blynk app (and it functions properly)
- [ ] Sound of melody playing

Deliverables

Submit on Brightspace (https://brightspace.usc.edu)

1. Fritzing layout file

2. A compressed file containing your project. Follow the guidelines for full credit.

   Here are the instructions for submission

   - Navigate to your project folder.
   - Include the *entire* folder in a zip file
   - Rename the zip file based on naming convention
   - Upload zip file to Brightspace site for our course

3. A photograph of your device connected to USB with the cyan light on (indicating it is connected).
4. ==A screenshot of your custom Blynk app==
5. ==A screenshot of your custom datastream table from website==
6. A video demonstrating your project functioning (either upload this to separate from the zip file, or upload to YouTube / Google Drive / Dropbox and provide the link in your submission).

Grading

Item	Points
Fritzing layout	5
Stage 1: Create State Machine States and Transition Logic	**30**
`enum` models	5
State transition logic / correct state changes	10
State duration logic / correct state durations	10
Magnetic switch holds the brewing state change	5
Stage 2: Connect Blynk, Set Up Datastreams, and Design Blynk App	**4**
Datastreams correct configured	2
Blynk user interface app elements	2
Stage 3: Enable communication between Blynk app and Photon 2	**6**
`Button Start Brew` starts and stops brewing process	2
`Button Sound` enables / disables sound playing	2
Stage 4: Enable communication between Photon 2 and app	**6**
`String State` changes with correct state	2
`String Cup Status` updates	2
Stage 5: Play Song	**4**
Song melody and duration is stored correctly and at least 16 notes	2
Song plays at end of **Pour** state	2
Total	51

Recommendation #1: Music Notes as Constants

- You can play all tones directly as frequencies for this assignment
- However, it can be helpful to create melodies with musical notes instead of frequencies.
- The following code file lists the musical notes as frequencies: pitches.h
- Download `pitches.h` to the `src` folder for your project.
- In your main `.ino` sketch, you can then include this file and use the constants to play musical notes.
- Example

```
#include "pitches.h"

// example
tone(D6, NOTE_D4, 500); //play the D note above middle C
```

Recommendation #2: Blynk Starting Template Code

- You can adapt the Blynk code we used in class, or you can start with this empty Blynk template. Make sure to install the `blynk` library
- blynk_start_template.cpp

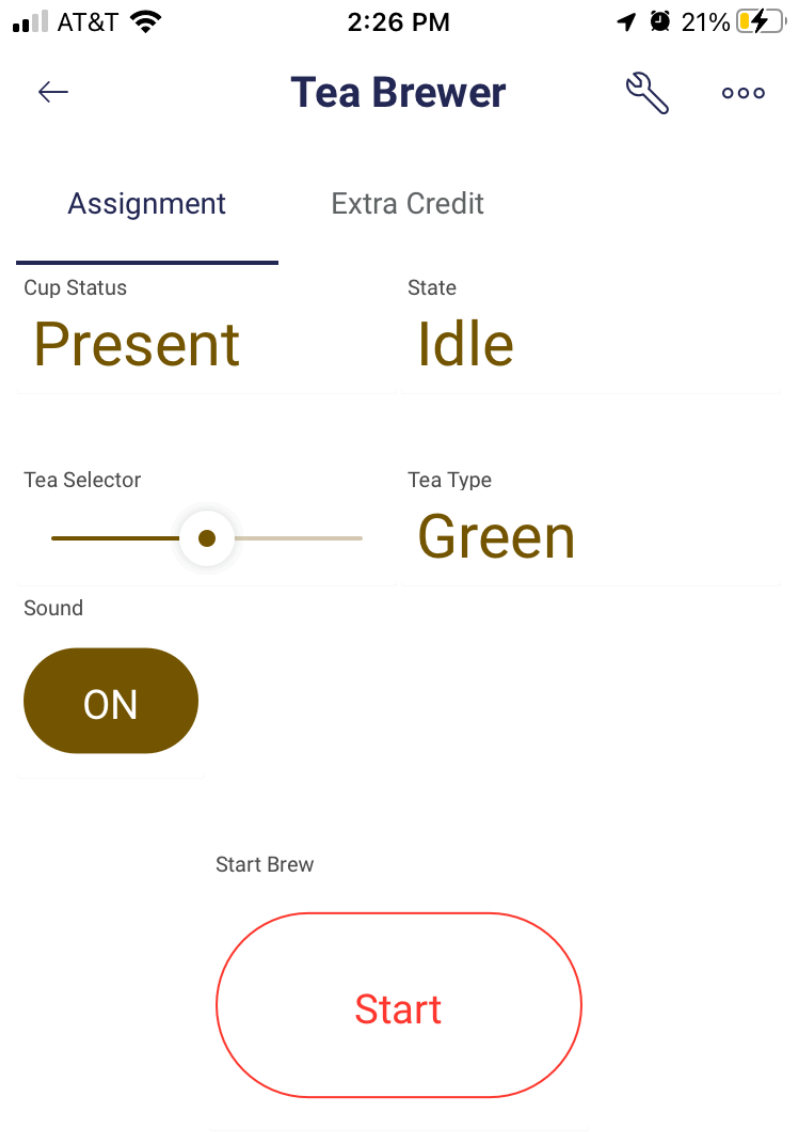
Extra Credit Idea #1: Tea Selector

- For extra credit, you will add the ability for the user to choose different types of tea. When the user changes a slider on the Blynk, it will change the length of the **Steep** stage

State Transition Table

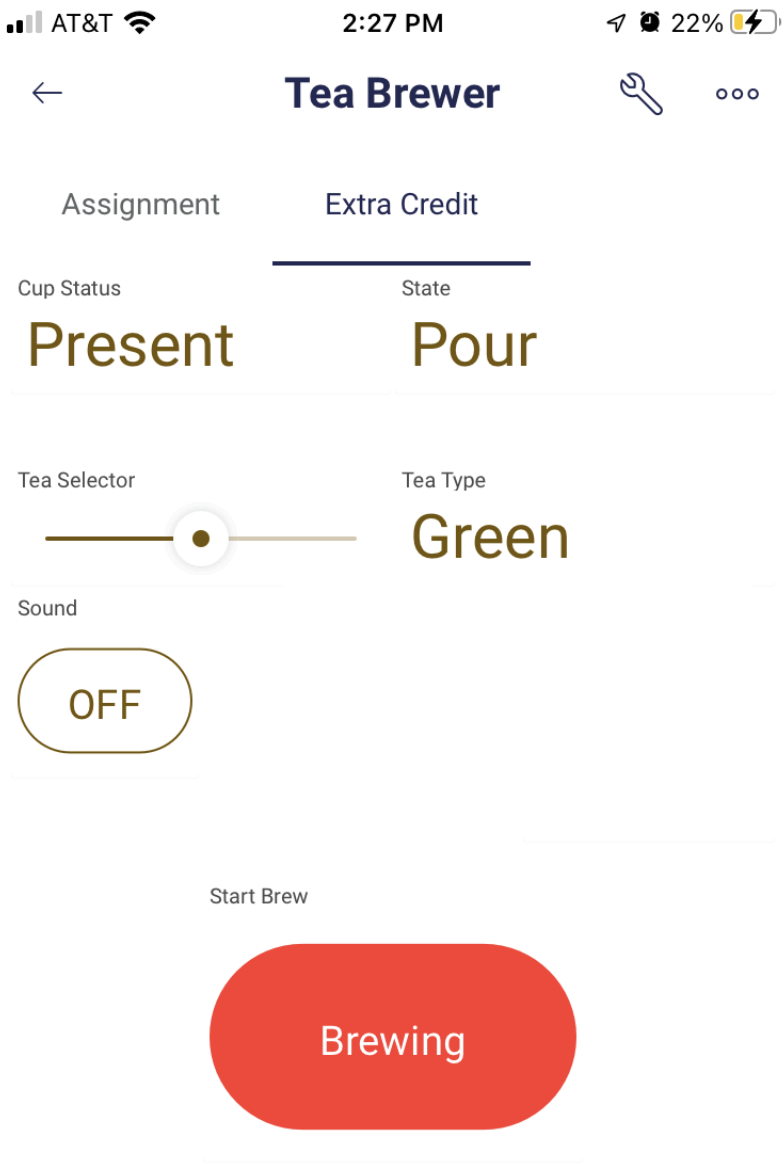
State	Tea Type	Length in Steep State
Steep	Green	2 seconds
Steep	Black	3 seconds
Steep	Herbal	4 seconds

Blynk App



Tea Type Selector and Tea Display

- When slider is moved,
  - Tea selection moves from 0, 1, or 2
  - Status should be updated with name of tea



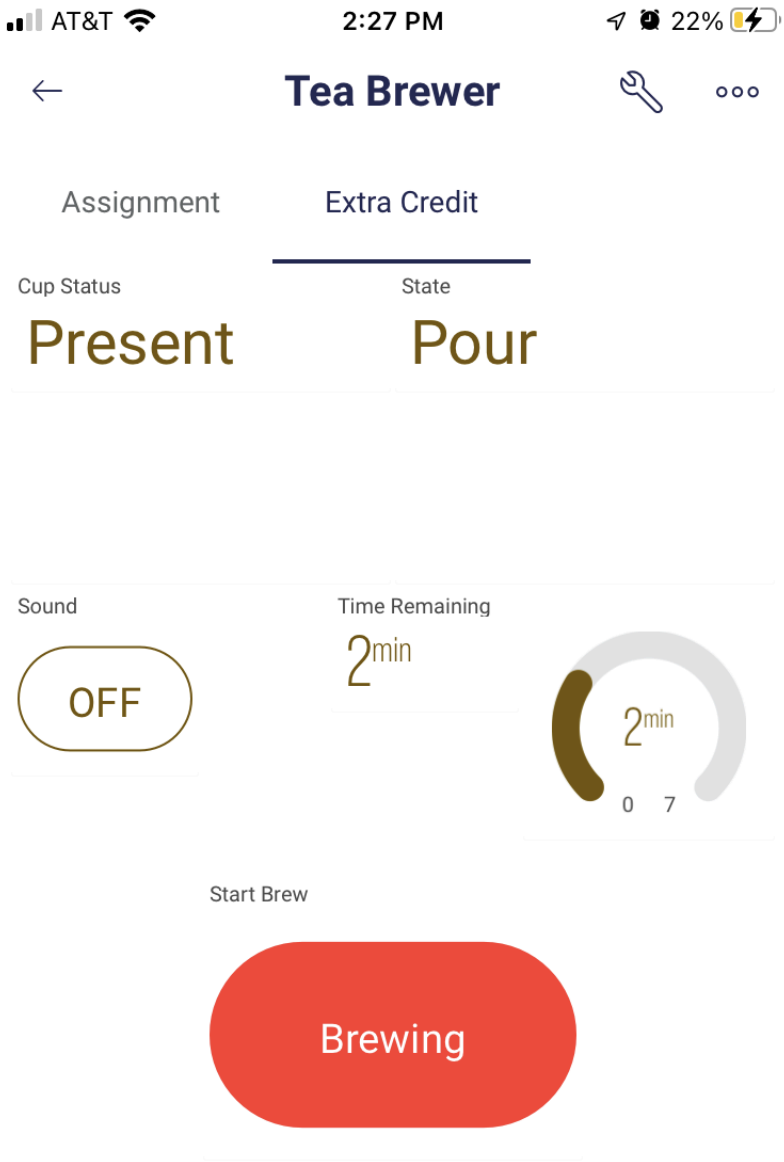
- To set this up, add two new app elements: value display and gauge

- Add two new datastreams

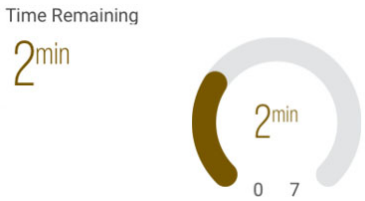
Name	Pin	Type	Values	Data FROM	Data TO
`Slider Tea Type`	`V0`	integer	`0` for black tea, `1` for green tea, `2` for herbal tea	app	Photon 2
`Text Tea Type`	`V1`	String	`Black`, `Green`, `Herbal`	Photon 2	app

Extra Credit Idea #2: Time Remaining

- For extra credit, you can add a countdown timer that shows the remaining time in each stage



- To set this up, add two new app elements: value display and gauge



- Add one new datastream

Name	Pin	Type	Values	Data FROM	Data TO
`Time Remaining`	`V6`	integer	`0` as min, `7` as max	Photon 2	app

- In your code, track the time remaining and send to app

Updated: February 14, 2026